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Faculty of Engineering, Computer Science and Psychology

Module Handbook

Master's degree program M.Sc. Artificial Intelligence for Connected Industries

Winter semester 2025/2026

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1 Artificial Intelligence and Machine Learning for Connected Systems

Token / Number:	AIML
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Eduard Garcia (UPC)
Training staff:	Eduard Garcia (UPC) David Remondo (UPC)
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Module,
Requirements (contentual):	Calculus, Algebra, basic concepts of statistics and probability. Prior knowledge on Python is strongly recommended
Learning objectives:	The main goal of this course is to cover the basic concepts related to machine learning projects and present the main ML models and algorithms and how to apply them to connected systems.
Content:	The course alternates between theoretical lectures and lab sessions. The main idea consists of presenting the theoretical background of a specific subject, followed by a lab session in which students will learn more details about each model and algorithm with practical examples using the most popular tools and libraries available. The course includes hands-on lab sessions with practical assignments, some of which are evaluated. Assessment is completed through a written exam and a final project presented as a competition after the lectures conclude. The course is connected-systems oriented, which means that, in addition to the most popular datasets, like MNIST and California houses, students will also see other examples of network-related datasets. **Topics:** - Introduction to AIML. - Practical skills and Linear Regression. Lab: end-to-end work, exploratory data analysis. - Supervised Learning and Classification (Decision Trees and Random Forest, Bayesian Detection, Non-Parametric Classifiers) Lab: Classification, Linear and Quadratic Discriminants, K-nearest neighbors (KNN). - Dimensionality Reduction Lab: Principal Component Analysis (PCA), Multiple Discriminant Analysis (MDA) - Unsupervised Learning Lab: Clustering - Artificial Neural Networks, Deep Neural Networks (DNN) Lab: Neural Networks, Multi-Layer Perceptron (MLP) - Training enhancement techniques (e.g. Ensembles, in DNN)

Content (continued):	**Complementary content:** - Basics on Reinforcement Learning Lab: Basics of Reinforcement Learning - Data processing tools (e.g. TensorFlow, Scikit learning) Lab: management of time-series in Recurrent Neural Networks (RNN) Lab tools - Language: Python - Frameworks and libraries: Numpy, Pandas, Matplotlib, Scikit-Learn, Tensor Flow, Keras, Google Colab, LaTeX, and Overleaf Datasets - Iris - Ridership: Bus and Rail rides in Chicago - CTU-13: 13 attack scenarios from botnets - 5G-Traffic: Traffic load in different cities in France - LiveStreaming: Live streaming data of user's connections (World Cup matches).
Literature:	- James, G., Witten, D., Hastie, T., Tibshirani, R. "An Introduction to Statistical Learning", 2nd edition, 2021 - Aurélien Géron, "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow", 2nd Edition, O'Reilly Media, 2019 - Tom Mitchell, "Machine Learning", McGraw-Hill Science, 1997 - R. O. Duda, P. E. Hart, D. G. Stork, Pattern Classification, Ed. Wiley Interscience, 2002. - Christopher M. Bishop, Pattern Recognition and Machine Learning, Springer, 2006
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 180 h
Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

2 Master Thesis

Token / Number:	MA
Credits:	30 ECTS
Language:	English
Turn / Duration:	every Semester / 1 Semester
Module authority:	Prof. Dr. Birte Glimm
Training staff:	Master's thesis supervisor
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Module, Master thesis
Requirements (contentual):	It is desirable to have taken basic modules in the field of the master's thesis.
Learning objectives:	Students graduating from the master's degree are expected to integrate the following sectors: - Smart factories, Industry 4.0. - Datacenter and cloud providers - IoT software editors - Embedded systems manufacturer - Internet and mobile application editors - Telecommunication network operators. - Smart-city and smart-grid network providers - Artificial intelligence start-ups - Security and defense
Content:	This unit is dedicated to the development of the master's specialisation dissertation on the definition, evaluation, modelling and/or experimentation of new algorithms, protocols, network architectures and IoT.
Literature:	Depending on the specific topic.
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Selection of an appropriate topic **Sum:** 900 h
Grading procedure:	Evaluation of written dissertation report and final defence.
Requirements (formal):	not specified

3 Design of Embedded Real-Time Systems

Token / Number:	EES
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Prof. Dr.-Ing. Frank Slomka
Training staff:	All lecturers of the Department of Engineering and Computer Science
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Systems and Networks
Requirements (contentual):	Fundamentals of computer architecture or embedded systems architecture
Learning objectives:	In taking part in the module "Design Methodology of Embedded Systems" the students learn how to describe and sketch the model-based design of embedded systems. They will be able to name and distinguish different analytic processes for the assessment of embedded systems. From a set of different methods and algorithms for the analysis of real-time operation, they will learn how to pick a suitable method in order to solve a given problem. The participants will be enabled to build new methods and algorithms and to prove their correctness. The students will be able to identify the complexity of algorithms and to develop approximations. Moreover, they will be able to assess various designs of embedded systems and to compare them.
Content:	Even if you cannot see them – embedded systems are everywhere. Very often, the only time that we take notice of them is when they no longer function properly: All of a sudden, the expensive new car refuses to go any further. Doors in the shopping mall do not open or close anymore at closing time. The barrier in the parking garage remains closed. The smart phone cannot login at the network. Not only can certain bugs take away comfortable functions, incorrectly designed systems can be extremely dangerous and become very expensive. For example, there were certain incidents when rockets that had expensive satellites on board had to be blown up because they lost their intended trajectories and became a threat. One of the most cost-intensive mistakes throughout the history of embedded systems was the explosion of an "Ariane" rocket in 1996. The damage reached 370 Million US-Dollar and was caused by an overflow of a register which on the other hand was the outcome of the fact that Ariane 5 accelerated quicker than the predecessor Ariane 4. Embedded Systems became so complex during the course of the last decades that methods of computer-assisted design have to be applied. This module deals with the building of models and the analysis of embedded systems, focusing on the design of a uniform model for event-driven real-time systems.
Content (continued):	The course focuses mainly on: - Model-based design of embedded systems - Time and real-time systems - Modeling of embedded systems: event models and graphs - Intrinsic analysis of real-time systems - Extrinsic analysis of real-time systems - Complexity and approximation of the extrinsic analysis - Optimization and Design Space Exploration

Literature:	- Jürgen Teich: Digitale Hardware-/Software Systeme, Springer 1996 - Peter Liggesmeyer und Dieter Rombach: Software Engineering eingebetteter Systeme, Spektrum Akademischer Verlag 2005 - Jean J. Labrosse: Embedded Systems Building Blocks, CMP 2000 - Peter Marwedel: Eingebette Systeme, Springer 2007 - Zbigniew Michalewicz und David B. Fogel: Modern Heuristics, Springer, 2000
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 180 h
Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

4 Management Aspects of Systems Engineering I

Token / Number:	SyE I
English title:	Management Aspects of Systems Engineering I
Credits:	6 ECTS
Language:	English or German
Turn / Duration:	every Summer Semester / 1 Semester
Module authority:	Michael Leute (Fa. Airbus Defence and Space)
Training staff:	Michael Leute Sascha Ackva
Integration of module into courses of studies:	Sensor Systems Engineering, M.Sc., Compulsory Elective Module, Management Aspects
Requirements (contentual):	None
Learning objectives:	Today, starting at household applications providing automation services up to highly complex systems as the International Space Station a lot of functionality is integrated and thus systems get more and more complex. In addition, the environment a system is installed imposes constraints increasing the complexity. Those constraints could be raised by a competitive market, by already installed systems or are given by laws and regulations. Systems Engineering has developed over the decades from a capability of a few experienced senior engineers to an acknowledged profession. Systems Engineering has proven to be a key enabling factor for companies to handle and manage complexity and to ensure effective communication among all stakeholders allowing the development of successful systems. The course provides the rationale and foundation for applying Systems Engineering and an introduction into the thinking of Systems Engineering. The students explain the difference of a system in comparison to its system elements. They describe the role of a systems engineer within a project organisation and his or her work areas. The students understand the importance of upfront analysis in order to avoid unplanned costs in later life cycle stages. They recognize that communication, e.g. with specialist disciplines, is a fundamental element of the SE tasks. They further analyse a system completely on a specific abstraction level considering the complete life cycle before advancing into deeper details towards implementation. The students work with a wide scope overlooking all technical aspects of the whole system instead of focusing solely on a single detail. The students elicit requirements starting from the stakeholder needs, identify possible solution architectures, perform trade studies and decompose and allocate the requirements down the system hierarchy. They apply different techniques to validate architecture, design and requirements. The students manage the technical aspects of a system and coordinate with the specialist engineering disciplines as well as project management.

Content:	- Definition of a system and systems engineering, system hierarchy and development lifecycle - Justification and rationale for the systems engineering discipline i.e. managing complexity, effective communication and common understanding - The role of the systems engineer - Systems engineering processes - Typical reviews (quality gates) in the development life cycle, and their objectives - Requirements engineering - System architecture - Integration and testing - Project life cycle and development models
Literature:	- INCOSE Systems Engineering Handbook (ISBN 978-1-937076-02-3) - Systems Engineering, Principles and Practice, Kossiakoff et.al.
Modes of learning and teaching:	**On-campus meetings** (e.g. kick-off, laboratories, exercises, group work, presentations, and exam): 16 h, distributed over 4 days**Distance learning:** **Synchronous** (e.g. webinar, group work, presentation, chat): 24 h **Asynchronous** (e.g. self-study based on lecture notes, videos, literature, exercises, e-learning assignments): 140 h
Estimation of effort:	Präsenzstudium, Synchrones und Asynchrones Selbststudium: 180 h
Grading procedure:	The grade of the module will be the grade of the exam.
Requirements (formal):	Requirement is a first academic degree.

5 Systemtechnik

Token / Number:	SyT
Credits:	6 ECTS
Language:	Deutsch
Turn / Duration:	every Summer Semester / 1 Semester
Module authority:	Prof. Dr.-Ing. Jian Xie
Training staff:	Prof. Dr.-Ing. Jian Xie
Integration of module into courses of studies:	Sensor Systems Engineering, M.Sc., Compulsory Elective Module, Fundamentals of Systems Engineering
Requirements (contentual):	- Berechnung von Gleichstrom- und Wechselstromkreisen - Grundkenntnisse über elektronische Bauelemente und Schaltungen
Learning objectives:	Der Erfolg eines Unternehmens hängt immer mehr von der Fähigkeit ab, komplexe und interdisziplinäre Projekte erfolgreich zu gestalten. Die Systemtechnik beschreibt Methoden, Strategien und Vorgehensweisen zur Abwicklung von solchen komplexen Vorhaben. Durch die Belegung dieses Moduls werden die Studierenden dazu befähigt, Projekte effizient und zielgerichtet zu planen, zu steuern und abzuwickeln. Die Studierenden können Grundbegriffe der Systemtechnik beschreiben, Denksätze darstellen und anwenden sowie alternative Vorgehensmodelle skizzieren. Sie lernen Betrachtungsweisen, Techniken und Vorgehensschritte für die Situationsanalyse, Zielformulierung, Synthese-Analyse sowie Bewertung und Entscheidung kennen und anwenden. Sie können Aufgaben und Inhalte verschiedener Projektphasen benennen und beschreiben sowie Projektorganisationen klassifizieren, vergleichen, analysieren und einsetzen, die Einsatzgebiete verschiedener Projektorganisationen darstellen und Hilfsmittel wie Organigramme, Netzpläne, Ressourcenpläne oder Fortschrittspläne anwenden.
Content:	- Kennenlernen von Denkweisen, Methoden und Hilfsmitteln der Systemtechnik - Philosophie der Systemtechnik mit Grundbegriffen und Anwendung - Vorgehensmodelle wie Top Down, Variantenbildung, Phasengliederung, Problemlösungszyklus - Systemgestaltung mit Verfahren wie Situationsanalyse, Zielformulierung, Synthese-Analyse, Bewertung und Entscheidung - Projektmanagement mit Schwerpunkten wie Projektphasen, Organisationen, Methoden und Hilfsmitteln - Gruppenübung zu einem Entwicklungsprojekt
Literature:	- Haberfellner/Nagel: Systems Engineering, Verlag Industrielle Organisation Zürich
Modes of learning and teaching:	**On-campus meetings** (e.g. kick-off, laboratories, exercises, group work, presentations, and exam): 11 h, verteilt über 4 Tage E-Learning: Webinar: 16 h Self-Study: 153 h
Estimation of effort:	Impartment of knowledge: 42 h Preparation, Follow-up, Exercises, Practice: 132 h Miscellaneous: 5 h Exam: 0,5-1 h **Sum:** 180 h
Grading procedure:	Die Modulnote ergibt sich aus der Modulprüfung.

Requirements (formal):	Voraussetzung ist ein erster Hochschulabschluss.

6 Wireless Sensor Networks

Token / Number:	WSN
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Summer Semester / 1 Semester
Module authority:	Prof. Dr. rer. nat. Frank Kargl
Training staff:	All lecturers of the Department of Engineering and Computer Science
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Systems and Networks
Requirements (contentual):	Fundamentals in computer networks, operating systems, and computer architectures. Programming skills in C or Python.
Learning objectives:	Students can explain the essential requirements and unique challenges of wireless ad-hoc networking; they can identify differences to classical wired networks and explain them. Given specific application requirements, students are able to design simple examples of wireless sensor network systems and construct and evaluate these systems. In particular, students are able to select existing routing and data dissemination strategies and adapt or extend them to suit specific scenarios. They can describe state of the art of wireless sensor network research and seminal research works. Using the acquired methods, students independently analyze new literature in sub-domains that were not covered in the lectures. They explain fundamental concepts of security and privacy protection in WSNs, select appropriate protection mechanisms, and integrate them in system architectures.
Content:	- Introduction to Wireless Sensor Networks (WSN): requirements, applications, and examples - Hardware and software platforms for WSNs - Link-layer technologies for WSN communication - Networking protocols and data dissemination strategies - WSN security and privacy - Simulative and experimental evaluation - Related domains (e.g., vehicular ad-hoc networks, delay-tolerant networks)
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 180 h
Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

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7 Modules from “Systems and Networks” from partner universities

Token / Number:	not specified
Credits:	depends on the modules you choose ECTS
Language:	English
Turn / Duration:	every Semester / 1 Semester
Module authority:	Partner Universities
Training staff:	depends on the modules you choose
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Systems and Networks
Requirements (contentual):	depends on the modules you choose
Learning objectives:	depends on the modules you choose
Content:	Recommended modules: - Parallel and Distributed Systems - Wireless and Mobile Networks - Network Virtualization and Automation
Literature:	Depending on the specific topic.
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	depends on the modules you choose
Grading procedure:	depends on the modules you choose
Requirements (formal):	not specified

8 Control System Theory and Control Engineering

Token / Number:	STCE
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Dr.-Ing. Sönke Rhein
Training staff:	All lecturers of the Department of Engineering and Computer Science
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Automatics and Robots
Requirements (contentual):	- Fundamentals of higher mathematics, especially linear algebra - Fundamentals of signals and systems - Basic design methods for LTI systems in frequency domain - Description of linear, time-invariant systems in time and frequency domain, Laplace transform, analysis of LTI systems (Bode and Nyquist plots)
Learning objectives:	To an increasing degreee, requirements on safety, sustainability and economic feasibility of technical products and production plants call for modern approaches of control-theory-based methods. Especially in the environment of control theory, simple and heuristically designed controllers have been reaching their limits. Systematic design of model-based controllers in time domain allows considering of non-linearities and has the potential to achieve significantly improved controller results. Students are able to describe the necessary mathematics and system-theory basics. They are capable of applying the methods of model-based control theory to linear and non-linear systems. They are able to describe and analyze time-continuous systems in time domain. They are able to categorize systems according to system-theoretic properties. Further, they have the ability to apply formal methods in order to design controllers in time domain. They are also able to apply methods for designing controllers for non-linear systems and are capable of managing its operation.
Content:	- Linear and non-linear time-continuous systems in state space - Linearization and general solution of linear differential equations of states - Structural properties of LTI systems in state space (stability, controllability, observability - Design of state controllers and state observers for linear systems - Analysis of non-linear systems (Lyapunov-stability) - Control and feedback control of non-linear systems
Literature:	- J. Lunze: Regelungstechnik 1: Systemtheoretische Grundlagen, Analyse und Entwurf einschleifiger Regelungen. 4 Auflage Springer-Verlag, Berlin, 2006 - J. Lunze: Regelungstechnik 2: Mehrgrößensysteme, Digitale Regelung. Springer-Verlag, Berlin, 2004 - T. Kailath: Linear Systems. Prentice Hall, Upper Saddle River, 1980 - W.J. Rugh: Linear System Theory, Prentice Hall, 1996 - H.K. Khalil. Nonlinear Systems. Prentice Hall, Upper Saddle River, 2002 - S. Sastry. Nonlinear Systems. Springer, New York, 1999 - A. Isidori. Nonlinear Control Systems. Springer, Berlin, 3rd edition, 1995
Basis for:	Module Modeling and identification of dynamic systems, master's thesis in the domain of control and automatization technology

Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 180 h
Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

9 Learning Robots

Token / Number:	LR
Credits:	3 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Prof. Dr. Birte Glimm
Training staff:	Lecturers from the Institute of AI, Ulm University
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Automatics and Robots
Requirements (contentual):	Programming in Python or C++, basics of ROS and simulation environments, understanding of machine learning concepts (e.g. Neural Nets, Reinforcement Learning, Computer Vision) may be helpful.
Learning objectives:	Students are able to conduct a project-oriented, scientific work in the area of robotics. They have the ability to define an innovative project topic, acquire skills and technologies required for the project, find related works and base their own work on them where appropriate. The students are further able to work independently in a team and to apply state-of-the-art methods to develop concepts and solutions for the project topic. They can document the results in the form of a scientific report and present their findings in the form of a presentation. They are able to acquaint themselves with a new topic and to conduct a scientific project.
Content:	At the beginning of the project, the range of contents that should be covered in the project will be discussed in detail and the relevant methods will be identified. This is accompanied by a literature search, reading, and discussion phase. In this phase relevant mathematical and computational methods will be selected and discussed. In the second phase, the team starts to implement the experimental design on a simulated robot. Regular team meetings and supervisory consultations lead to an iterated improvement of the software. In the third phase, the thoroughly tested software will be transferred to the robot and tested in a real or simulated environment. The results are written up in a final project report and are presented in a final project presentation. - Definition of concrete project idea. - Project plan incl. systematic literature review. - Evaluation of suitable technologies. - Self-learning of the required technical foundations. - Architecture design. - Implementation. - Integration. - Test. - Deployment.
Literature:	- Thrun, Burgard, Fox "Probabilistic Robotics" - Bishop "Pattern recognition and machine learning" - Russell and Norvig "Artificial intelligence. A modern approach" - Goebel "ROS by Example INDIGO – Volume 1"
Modes of learning and teaching:	Blended Learning Format

Estimation of effort:	Blended Learning: 90 h
Grading procedure:	Assessment (Project Report): The graded assessment consists of a project report that documents the planning, execution, and results of the project. The module grade is equal to the examination grade(s).
Requirements (formal):	Participation in the LR module requires successful completion of the Robot Operating System (ROS) module.

10 Modeling and Identification of Dynamic Systems

Token / Number:	MIDS
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Dr.-Ing. Sönke Rhein
Training staff:	All lecturers of the Department of Engineering and Computer Science
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Automatics and Robots
Requirements (contentual):	- Fundamentals of higher mathematics, especially linear algebra - Fundamentals of signals and systems - Basics of control theory in frequency and time domain
Learning objectives:	Appropriate descriptions of systems are the base of many methods in control theory and are required for model-based monitoring. Model-based techniques open extensive paths to optimize existing control systems in industrial applications. The main requirement are proper mathematical models which on the one hand represent all important dynamic effects as precisely as possible, and stay feasible regarding computational complexity on the other hand. It is further important to be able to determine parameters which are not observable. Students are able to describe and apply methods of mathematical modeling of technical processes based on physical principles. They are further able to describe technical systems of various physical domains using mathematical formalisms. They are especially able to derive suitable models for controller design and to parameterize the models with identification procedures, e.g. with using black box models. Students are capable of designing optimal state estimaters and state controllers. Thereby, they can apply according methods of identification, estimation, and control.
Content:	- Modeling mechanical, electrical, and hydraulic systems - Parameter-based and non-parameter-based identification approaches - Optimal estimation procedures and filter (e.g. Kalman filter)
Literature:	- P.E. Wellstead: Physical Systems Modelling, Academic Press, 1979 - R. Isermann: Mechatronische Syteme: Grundlagen, Springer, 2002 - R. Isermann: Identifikation dynamischer Systeme 1 und 2, Springer, 1992 - D.G. Luenberger: Optimization by Vector Space Methods, John Wiley & Sons, 1969 - A. Gelb: Applied Optimal Estimation, M.I.T. Press, 1974 - A.E. Bryson, Y.-C. Ho: Applied Optimal Control, Hemisphere Publishing Corporation, 1975
Basis for:	Master's thesis in the domain of control and automatization technology
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 180 h

Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

11 Pattern Recognition and Deep Learning

Token / Number:	PRDL
German title:	Mustererkennung und mehrschichtiges Lernen
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	apl. Prof. Dr. Friedhelm Schwenker
Training staff:	apl. Prof. Dr. Friedhelm Schwenker
Integration of module into courses of studies:	Sensor Systems Engineering, M.Sc., Compulsory Elective Module, Fundamentals of Systems Engineering
Requirements (contentual):	- Basic knowledge in programming and basic concepts of analysis, linear algebra, and probability.
Learning objectives:	Students acquire knowledge about different methods and algorithms of pattern recognition and deep artificial neural networks. In exercises, students are able to implement the basic algorithms, will apply pattern recognition principles to technical applications, and learn how to evaluate the performance of classifiers.
Content:	In this course the basic topics on statistical pattern recognition and deep neural networks are introduced: - Introduction to statistical and neural pattern recognition - Linear and nonlinear classifiers - Kernel methods and deep neural network learning - Feature extraction, selection and reduction - Applications and system performance evaluation
Literature:	- Bishop, Chris: Pattern Recognition and Machine Learning, Springer, 2007 - Theodoridis, Sergios & Koutroubas, Konstantionos, Pattern Recognition, Academic Press, 2010 - Charu Aggarwal: Neural Networks and Deep Learning, Springer, 2018
Modes of learning and teaching:	**On-campus meetings** (e.g. kick-off, laboratories, exercises, group work, presentations, and exam): 8 h E-Learning: Webinar: 28 h Self-Study: 144 h
Estimation of effort:	Impartment of knowledge: 36 h Preparation, Follow-up, Exercises, Practice: 138 h Miscellaneous: 5 h Exam: 1 h **Sum:** 180 h
Grading procedure:	Grade of the module will be the grade of the exam.
Requirements (formal):	Requirement is a first academic degree.

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12 Robot Operating Systems

Token / Number:	
Credits:	3 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Prof. Dr. Birte Glimm
Training staff:	Lecturers from the Institute of AI, Ulm University
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Automatics and Robots
Requirements (contentual):	Linear algebra and analysis, programming in Python or C++
Learning objectives:	Students will understand the fundamental theoretical and algorithmic principles behind robotic systems. Students are able to solve robot specific learning problems involving, for example, navigation and mapping, grasping and manipulation, and interaction with humans. They understand the ROS Ecosystem (topics, nodes, messages, services, actionlib) and are able to develop simple applications to control robot motion.
Content:	This course gives an introduction to the Robot Operating System (ROS) including available tools that are commonly used in robotics. With the help of examples, the course provides a starting point for working with robots. The course covers how to create software including simulation, to interface sensors and actuators, and to integrate control algorithms. **Topics:** - ROS architecture: Master, nodes, topics, messages, services, parameters and actions. - Console commands: Navigating and analyzing the ROS system and the catkin workspace. - Creating ROS packages: Structure, launch-files, and best practices. - Simulating with ROS: robot models (URDF) and simulation environments. - Working with visualizations and user interface tools. - Introduction to ROS2.
Literature:	- Thrun, Burgard, Fox "Probabilistic Robotics" - Bishop "Pattern recognition and machine learning" - Russell & Norvig "Artificial intelligence. A modern approach" - Goebel "ROS by Example INDIGO – Volume 1"
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 90 h

Grading procedure:	Portfolio Examination (Quizzes and Exercises) The portfolio examination consists of two components: Quizzes: Short online tests to verify understanding of fundamental ROS concepts Exercises: Practical programming tasks focusing on the application of ROS functionalities All tasks (quizzes and exercises) in the course must be passed; the final grade is determined by the weighted total score. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

13 Signals and Systems

Token / Number:	SaS
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Dr. Werner Teich
Training staff:	All lecturers of the Department of Engineering and Computer Science
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Automatics and Robots
Requirements (contentual):	Advanced Mathematics
Learning objectives:	The concepts of signals and systems are powerful tools for any engineer dealing with information bearing, measurable physical quantities. Areas of applications include, among others, communications engineering, signal processing, control engineering, and systems engineering. The students will be able to classify, interpret, and compare signals and systems with respect to their characteristic properties. They can explain and apply analytical and numerical methods to analyze and synthesize signals and systems in time and frequency domain. Suitable signal transformations can be chosen and calculated with the help of transformation tables. The students are able to recognize stochastic signals and analyze them based on their characteristic properties. They can calculate and interpret the influence of linear time-invariant systems on stochastic signals.
Content:	- Basic properties of discrete-time and continuous-time systems - Z-transform - Basic properties of discrete-time and continuous-time systems - Linear time-invariant systems, convolution integral - Fourier transform, discrete Fourier transform, Fourier series - Sampling theorem - Probability theory, random variables, and stochastic processes - Stochastic signals and linear time-invariant systems
Literature:	- Alan V. Oppenheim and Alan S. Willsky: Signals and Systems, Prentice Hall 1996 - Mrinal Mandal and Amir Asif: Continuous and Discrete Time Signals and Systems, Cambridge University Press, 2007 - Athanasios Papoulis and S. Unnikrishna Pillai: Probability, Random Variables, and Stochastic Processes, McGraw-Hill, 2002 - Thomas Frey und Martin Bossert: Signal- und Systemtheorie, B.G. Teubner Verlag, 2004 - Jens Ohm und Hans Dieter Lüke: Signalübertragung, Springer Verlag 2010
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 180 h

Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

14 Modules from “Automatics and Robots” from partner universities

Token / Number:	not specified
Credits:	depends on the modules you choose ECTS
Language:	English
Turn / Duration:	every Semester / 1 Semester
Module authority:	Partner Universities
Training staff:	depends on the modules you choose
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Automatics and Robots
Requirements (contentual):	depends on the modules you choose
Learning objectives:	depends on the modules you choose
Content:	Recommended modules: - Robot Predictive Maintenance
Literature:	Depending on the specific topic.
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	depends on the modules you choose
Grading procedure:	depends on the modules you choose
Requirements (formal):	not specified

15 Big (Social) Data Analytics – Methoden und Anwendungen

Token / Number:	BSDA
English title:	Big (Social) Data Analytics – Methods and Applications
Credits:	6 ECTS
Language:	Deutsch
Turn / Duration:	every Summer Semester / 1 Semester
Module authority:	Prof. Dr. Mathias Klier
Training staff:	Prof. Dr. Mathias Klier
Integration of module into courses of studies:	Business Analytics, M.Sc., Compulsory Elective Module
Requirements (contentual):	- Grundlagen in Stochastik und angewandter Statistik, wie sie im Modul „Stochastische Modellierung und Simulation“ vermittelt werden
Learning objectives:	Unternehmen stehen heutzutage – bspw. über soziale Medien und das Internet (z.B. Online Social Networks, Wikis, Bewertungs- und Rezensions-Communities, Diskussionsforen), aber auch in traditionellen Datenbanken (z.B. Data-Warehouse, Kundendatenbanken) – enorme Datenmengen in strukturierter (bspw. Beziehungen zwischen Netzwerkakteuren) oder unstrukturierter Form (bspw. Textinhalt von Tweets) zur Verfügung. Die zielgerichtete, fundierte und ethisch einwandfreie Analyse dieser Daten mittels automatisierter Verfahren aus den Bereichen Social Network Analysis und Big Data Analytics (z.B. Text Mining) ermöglicht eine verbesserte Entscheidungsunterstützung und birgt großes Potenzial, bspw. in den Bereichen Produktentwicklung, Marketing und Customer Relationship Management. Im Modul „Big (Social) Data Analytics – Methoden und Anwendungen“ werden hierfür zentrale Methoden und Techniken wie z.B. Community Detection in sozialen Netzwerken, Sentimentanalyse von Kundenstimmen und Klassifikation von Tweets mittels neuronaler Netze vermittelt und fallstudienbasiert mit realen Daten und Cases mit der Programmiersprache Python illustriert. Studierende des Moduls sind in der Lage, diese Methoden und Techniken fachgerecht auf betriebswirtschaftliche Fragestellungen anzuwenden, erkennen Probleme und Risiken bei der Anwendung (Robustheit sowie Notwendigkeit menschlicher Aufsicht) und können diesen mit entsprechenden Maßnahmen begegnen (Sicherstellen der Datenqualität und Herstellen von Transparenz mit Methoden der erklärbaren KI).
Content:	- Einführung: Big (Social) Data Analytics als hoch relevantes Thema - Methoden und Anwendungen der Big Data Analytics am Beispiel Social Media Analytics - Social Media Analytics – Soziale Netzwerkanalyse (z. B. Identifizierung einflussreicher Nutzer mithilfe von Vernetzungsmaßen, Community Detection) - Social Media Analytics – Text Mining (z.B. Klassifikation von Texten mit Machine Learning Methoden, Sentimentanalyse) - Zusammenfassung und kritische Würdigung

Literature:	- Heyer, G., Quasthoff, U., Wittig, T. (2006), Text Mining: Wissensrohstoff Text: Konzepte, Algorithmen, Ergebnisse. Bochum: W3L-Verlag. - Krippendorff, K.-H. (2012), Content Analysis: An Introduction to its Methodology, London: Sage Publications. - Leskovec, J., Rajaraman, A., Ullman, J.-D. (2014), Mining of Massive Datasets, Cambridge: Cambridge University Press. - Liu, B. (2011), Web Data Mining: Exploring Hyperlinks, Contents, and Usage Data, Heidelberg: Springer. - Wasserman, S., Faust, K. (1994), Social Network Analysis: Methods and Applications, Cambridge: Cambridge University Press. - Werner, A. (2013), Social Media – Analytics und Monitoring: Verfahren und Werkzeuge zur Optimierung des ROI, Heidelberg: dpunkt.verlag. - Zafarani, R., Abbasi, M.-A., Liu, H. (2014), Social Media Mining: An Introduction, Cambridge: Cambridge University Press.
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Impartment of knowledge: 96 h Deepening exercises: 76 h Online meeting: 6 h Exam: 2 h**Sum:** 180 h
Grading procedure:	Die Modulnote ergibt sich aus dem Ergebnis der Modulprüfung.
Requirements (formal):	keine

16 Business Process Management

Token / Number:	BPM
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Prof. Dr. Manfred Reichert
Training staff:	All lecturers of the Department of Engineering and Computer Science
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Processes and Data Management
Requirements (contentual):	Basic knowledge of databases, as taught in the module "Fundamentals of database systems".
Learning objectives:	After successfully completing the module, students are able to analyze, model, and optimize business processes on a professional level. They can apply the available methods, concepts, and software tools. Furthermore, they can explain how business processes can be supported by information systems and are able to identify the requirements for implementing such process-oriented information systems. Participants can also describe the essential characteristics, components, and functions of process-oriented information systems and place them within an overall architecture. In addition, they are capable of describing and applying various paradigms for modeling and developing process-oriented information systems, as well as evaluating their advantages and disadvantages. Finally, participants are equipped to implement selected process scenarios using a process management system.
Content:	- Introduction to business process management and case studies - Characteristics of process-oriented information systems - Analysis and optimization of business processes - Tools, languages, and guidelines for process modeling (e.g., Business Process Modeling Notation 2.0) - Modeling and verification of executable processes (i.e., workflows) - Implementation and execution of processes using process management technology - Selected architectural and implementation aspects of process management systems - Concepts and technologies for supporting flexible processes - Current trends in business process management
Literature:	- Lecture script and exercise materials - Reichert, M., Weber, B. (2012), Enabling Flexibility in Process-Aware Information Systems – Challenges, Methods, Technologies: Springer. - Weske, M. (2012), Business Process Management: Concepts, Languages, Architectures, 2. Auflage: Springer. - Dumas, M., La Rosa, M., Mendling, J., Reijers, H. (2013), Fundamentals of Business Process Management: Springer.
Modes of learning and teaching:	Blended Learning Format

Estimation of effort:	Blended Learning: 180 h
Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

17 Data and Process Mining

Token / Number:	DPM
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Summer Semester / 1 Semester
Module authority:	Prof. Dr. Manfred Reichert
Training staff:	Prof. Dr. Manfred Reichert
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Processes and Data Management
Requirements (contentual):	Algorithmics, data structures, statistics
Learning objectives:	The course aims to present data mining and knowledge discovery concepts, methods and techniques.
Content:	The students will learn various data analysis techniques and will apply these techniques for solving data mining problems using special software systems and tools. Topics: - Introduction - Concept description and definitions - Data preparation - Discovering, ingesting, and exploring data - Transforming data into analytics-ready data - Association rules - Clustering - Classification - Data mining - Model assessment and validation Complementary content: - Network analysis - Process Mining: Event Logs, Process Discovery, Conformance Checking, Log-based Verification (Toolset: ProM, Disco and Celonis) - Data Warehousing: ETL Process, Data Warehouse Components and Architecture, Multi-dimensional data model, ROLAPS

Literature:	- S. Chakrabarti et al, Data Mining. Know It All, Morgan Kaufmann, 2009 - K. Cios, W. Pedrycz, R. Swiniarski, L. Kurgan, Data Mining. A Knowledge Discovery Approach, Springer, 2007 - J. Han, M. Kamber, Data Mining: Concepts and Techniques, 2nd Edition, Morgan Kaufmann, 2006 - P. Tan, M. Steinbach, V. Kumar, Introduction to Data Mining, Addison Wesley, 2006 - D. Larose, Discovering Knowledge in Data. An Introduction to Data Mining, John Wiley and Sons, 2005 - Han, J., Kamber, M., Data Mining: Concepts and Techniques, 1st Edition, Morgan Kaufmann, 2000 - Weka system and documentation (http://www.cs.waikato.ac.nz/ml/weka/) - A. Géron. Hands-on machine learning with scikit-learn and tensorflow: concepts, tools, and techniques to build intelligent systems. Sebastopol, CA: O'Reilly Media, Inc, 2017. ISBN 9781491962299 - H. Mohanty, P. Bhuyan, D. Chenthati, Deepak. Big Data: A Primer. New Delhi: Springer India, 2015, ISBN 9788132224945 - J. Leskovec, A. Rajaraman, J.D. Ullman. Mining of massive datasets, 2nd ed. New York, N.Y.; Cambridge University Press, 2014. ISBN 9781107077232 - R. Garreta, G. Moncecchi, Guillermo. Learning scikit-learn: machine learning in Python. Birmingham: Packt Publishing, 2013. ISBN 978178328193
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 180 h
Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

18 Numerische Methoden für Data Science

Token / Number:	NMDS
English title:	Numerical Methods for Data Science
Credits:	6 ECTS
Language:	Deutsch
Turn / Duration:	every Winter Semester / 1 Semester
Module authority:	Prof. Dr. Karsten Urban
Training staff:	Prof. Dr. Karsten Urban Prof. Dr. Stefan Funken
Integration of module into courses of studies:	Business Analytics, M.Sc., Compulsory Elective Module
Requirements (contentual):	- Mathematische Kenntnisse auf Bachelor-Niveau, insbesondere Lineare Algebra (Matrizen, lineare Gleichungssysteme, Eigenwerte und -vektoren, Normen, Skalarprodukte) und Analysis (Funktionen mehrerer Veränderlicher, Satz von Taylor)
Learning objectives:	Durch das Wachstum an Datenvolumen stehen Unternehmen vor der Herausforderung, diese extrem großen Datenmengen („Big Data“) speichern und analysieren zu können. Dies ist entscheidend, um sie für weitere Prozesse zu verwenden. Herkömmliche algorithmische Methoden, die alle Daten betrachten, sind entweder nicht mehr anwendbar oder benötigen zu lange Rechenzeiten. In diesem Modul werden Sie numerische Methoden, Verfahren und Algorithmen kennenlernen, die auch für große Datenmengen noch effizient arbeiten und so erlauben, aus größeren Datenmengen Muster zu erkennen und wichtige Informationen zu extrahieren. Sie werden diese Algorithmen u.a. in Python implementieren. Dieses Modul ist die Basis für die Konstruktion und Implementierung von Algorithmen für maschinelles Lernen.
Content:	- Numerische Lineare Algebra für Big Data-Anwendungen: Lösung linearer Gleichungssysteme, Singulärwertzerlegung - Numerische Lösung hochdimensionaler nichtlinearer Gleichungssysteme - Numerische Verfahren für Kalibrierungsprobleme - Numerische Verfahren für restringierte und unrestringierte Optimierung - Standard-Software für derartige Problemstellungen
Literature:	- Judd, K. (1999), Numerical Methods in Economics (Scientific and Engineering): MIT Press. - Beylkin, G., Mohlenkamp, M.-J. (2005), Algorithms for Numerical Analysis in High Dimensions, SIAM J. Sci. Comput., 26:2133–2159. - Quarteroni, A., Sacco, F., Saleri, F. (2002): Numerische Mathematik 1: Springer. - Quarteroni, A., Sacco, F., Saleri, F. (2002): Numerische Mathematik 2: Springer. - Deuflhard, P., Hohmann, A. (2008): Numerische Mathematik 1: de Gruyter. - Deuflhard, P., Bornemann, F. (2013), Numerische Mathematik 2: de Gruyter. - Hanke-Bourgeois, M. (2003): Grundlagen der Numerischen Mathematik und des Wissenschaftlichen Rechnens: Teubner. - Bollhöfer, M., Mehrmann, V. (2004): Numerische Mathematik: Vieweg.

Modes of learning and teaching:	**On-campus meetings** (e.g. kick-off, laboratories, exercises, group work, presentations, and exam): 28 h E-Learning: Webinar: 96 h Self-Study: 56 h
Estimation of effort:	Impartment of knowledge: 96 h Deepening exercises: 76 h Online meeting: 6 h Exam: 2 h **Sum:** 180 h
Grading procedure:	Die Modulnote ergibt sich aus dem Ergebnis der Modulprüfung.
Requirements (formal):	keine

19 Project Management - Processes, Activities and Practices

Token / Number:	PMP
Credits:	6 ECTS
Language:	English
Turn / Duration:	every Summer Semester / 1 Semester
Module authority:	Dr. Volker Kraus
Training staff:	All lecturers of the Department of Engineering and Computer Science
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Processes and Data Management
Requirements (contentual):	not specified
Learning objectives:	Students who have successfully completed this module - are familiar with the basics of operational project management. - can independently plan, realize, monitor, and control complex interdisciplinary tasks. - are familiar with the various organizational forms of project management, the coordination of work in project teams, and the requirements and tasks of a project manager. - are proficient in the basic planning techniques of project management - are able to use different methods for planning, controlling, and monitoring processes based on network planning technology. - know the challenges of project management by means of practical examples.
Content:	- Ethics and Compliance. - Motivation, concept formation and basic elements of project management. - Business case and project selection criteria. - Project environment within an organization. - The role of the project manager in the company. - Project phases: initiate, plan, execute, monitor and close. - Scenarios and exercises to enhance knowledge transfer for selected topics. - Knowledge Areas: ◦ Stakeholder Management ◦ Integration Management ◦ Scope management ◦ Scheduling Management ◦ Cost Management ◦ Quality Management ◦ Resource management ◦ Communication Management ◦ Risk Management ◦ Procurement Management
Literature:	- A guide to the project management body of knowledge (PMBOK guide) / Project Management Institute. Sixth edition. Newtown Square, PA; Project Management Institute, 2017. ISBN: 978-1-62825-184-5 - Project Management Institute: Code of Ethics and Professional Conduct. - PMI Practice guides and selected additional literature for the relevant knowledge areas (will be cited during lectures).

Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	Blended Learning: 180 h
Grading procedure:	The module examination consists of one or more graded written or oral examination(s), depending on the number of participants. The examination(s) form will be announced in good time before the examination(s) is held – at least 4 weeks before the examination date. The module grade is equal to the examination grade(s).
Requirements (formal):	not specified

20 Modules from “Processes and Data Management” from partner universities

Token / Number:	not specified
Credits:	depends on the modules you choose ECTS
Language:	English
Turn / Duration:	every Semester / 1 Semester
Module authority:	Partner Universities
Training staff:	depends on the modules you choose
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Compulsory Elective Module, Processes and Data Management
Requirements (contentual):	depends on the modules you choose
Learning objectives:	depends on the modules you choose
Content:	Recommended modules: - Data Management and Digital Transformation in Industrial Process Automation
Literature:	Depending on the specific topic.
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	depends on the modules you choose
Grading procedure:	depends on the modules you choose
Requirements (formal):	not specified

21 Free Elective Modules from the Master's Degree Program Sensor Systems Engineering

Token / Number:	not specified
Credits:	ECTS
Language:	English
Turn / Duration:	every Semester / 1 Semester
Module authority:	Prof. Dr. Frank Slomka
Training staff:	depends on the modules you choose
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Free Elective Module,
Requirements (contentual):	depends on the modules you choose
Learning objectives:	depends on the modules you choose
Content:	Choose a module from the module catalog of Sensorsystemtechnik (Sensor System Engineering) at Ulm University. Recommended modules: - Pattern Recognition and Deep Learning - Management-Aspects of Systems Engineering I - Management-Aspects of Systems Engineering II
Literature:	Depending on the specific topic.
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	depends on the modules you choose
Grading procedure:	depends on the modules you choose
Requirements (formal):	not specified

22 Free Elective Modules from the Master's Degree Program Business Analytics

Token / Number:	not specified
Credits:	ECTS
Language:	English
Turn / Duration:	every Semester / 1 Semester
Module authority:	Prof. Dr. Mischa Seiter
Training staff:	depends on the modules you choose
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Free Elective Module,
Requirements (contentual):	depends on the modules you choose
Learning objectives:	depends on the modules you choose
Content:	Choose a module from the module catalog of Business Analytics at Ulm University. Recommended modules: - Machine Learning with Python - Advanced Python Programming
Literature:	Depending on the specific topic.
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	depends on the modules you choose
Grading procedure:	depends on the modules you choose
Requirements (formal):	not specified

23 Free Elective Modules from partner universities

Token / Number:	not specified
Credits:	ECTS
Language:	English
Turn / Duration:	every Semester / 1 Semester
Module authority:	partner universities
Training staff:	depends on the modules you choose
Integration of module into courses of studies:	Artificial Intelligence for Connected Industries, M.Sc., Free Elective Module,
Requirements (contentual):	depends on the modules you choose
Learning objectives:	depends on the modules you choose
Content:	Choose a module from the module catalog of the partner universities. Recommended modules: - Programming and Communication of a Robotic Arm - Refresh in C & Bash Programming - Network architecture - Complex Networks: Data Analysis and Network Science - Federated and Distributed Learning - Operations Research - Integration of Virtual and Augmented Reality Technologies in Connected Industries - Peer-to-Peer Systems and Blockchain - Datacenter Design and Operations - Contemporary Economic Issues / Company Organization - Ethics and Sovereignty of Digital Infrastructures - Industrial Internet of Things - Algorithm Engineering - Networks – Complements and Applications - Packet Switching and Processing Architectures - Sustainable IoT Technologies
Literature:	Depending on the specific topic.
Modes of learning and teaching:	Blended Learning Format
Estimation of effort:	depends on the modules you choose
Grading procedure:	depends on the modules you choose
Requirements (formal):	not specified

Based on Rev. **-2**. Last changed at **00.00.0000**, **00:00** from .